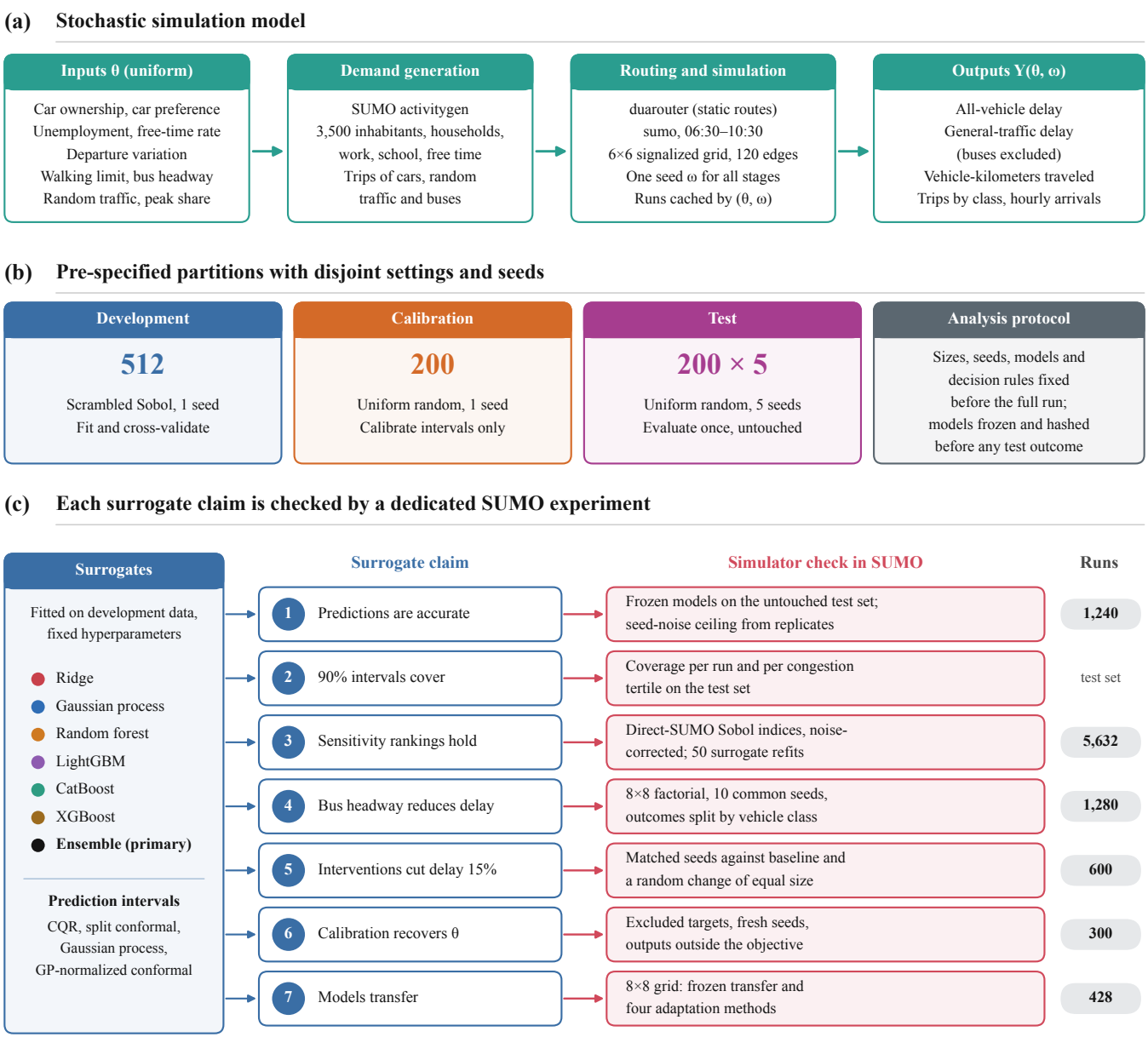

(a) Stochastic simulation model
Inputs θ (uniform)
Car ownership, car preference
Unemployment, free-time rate
Departure variation
Walking limit, bus headway
Random traffic, peak share
Demand generation
SUMO activitygen
3,500 inhabitants, households,
work, school, free time
Trips of cars, random
traffic and buses
Routing and simulation
duarouter (static routes)
sumo, 06:30–10:30
6×6 signalized grid, 120 edges
One seed ω for all stages
Runs cached by (θ, ω)
Outputs Y(θ, ω)
All-vehicle delay
General-traffic delay
(buses excluded)
Vehicle-kilometers traveled
Trips by class, hourly arrivals
(b) Pre-specified partitions with disjoint settings and seeds
Development
512
Scrambled Sobol, 1 seed
Fit and cross-validate
Calibration
200
Uniform random, 1 seed
Calibrate intervals only
Test
200 × 5
Uniform random, 5 seeds
Evaluate once, untouched
Analysis protocol
Sizes, seeds, models and
decision rules fixed
before the full run;
models frozen and hashed
before any test outcome
(c) Each surrogate claim is checked by a dedicated SUMO experiment
Surrogates
Fitted on development data,
fixed hyperparameters
Ridge
Gaussian process
Random forest
LightGBM
CatBoost
XGBoost
Ensemble (primary)
Prediction intervals
CQR, split conformal,
Gaussian process,
GP-normalized conformal
Surrogate claim
Simulator check in SUMO
Runs
1 Predictions are accurate
Frozen models on the untouched test set;
seed-noise ceiling from replicates
1,240
2 90% intervals cover
Coverage per run and per congestion
tertile on the test set
test set
3 Sensitivity rankings hold
Direct-SUMO Sobol indices, noise-
corrected; 50 surrogate refits
5,632
4 Bus headway reduces delay
8×8 factorial, 10 common seeds,
outcomes split by vehicle class
1,280
5 Interventions cut delay 15%
Matched seeds against baseline and
a random change of equal size
600
6 Calibration recovers θ
Excluded targets, fresh seeds,
outputs outside the objective
300
7 Models transfer
8×8 grid: frozen transfer and
four adaptation methods
428